Selected multimodal and layout-aware document models

First public paper year

	2021	2022	2023	2024	2025
Multimodal / layout foundations	VinVL LayoutXLM FLAVA	UDOP	KOSMOS-2 DocLLM (text + boxes)	Qwen2-VL	Qwen2.5-VL Qwen3-VL
High-resolution / efficient scale			Qwen-VL Monkey	mPLUG-DocOwl 1.5 TinyChart-3B MiniCPM-V InternVL 2.5	
OCR-free / visual parsing	Donut	Pix2Struct		mPLUG-DocOwl2	olmOCR SmolDocling DocSLM
Document-level models / agents					Docopilot CogDoc

Years follow official paper histories, not conference dates. Groups overlap; chronology does not imply replacement or performance superiority.
DocLLM uses text and bounding boxes without an image encoder. The final row includes native document models as well as agentic systems.